

Semi-supervised Medical Image Segmentation through Dual-task Consistency

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Abstract

Deep learning-based semi-supervised learning (SSL) algorithms have led to promising results in medical images segmentation and can alleviate doctors’ expensive annotations by leveraging unlabeled data. However, most of the existing SSL algorithms in the literature tend to regularize the model training by perturbing networks and/or data. Observing that multi/dual-task learning attends to various levels of information which have inherent prediction perturbation, we ask the question in this work: can we explicitly build task-level regularization rather than implicitly constructing networks-and/or data-level perturbation and then regularization for SSL? To answer this question, we propose a novel dual-task-consistency semi-supervised framework for the first time. Concretely, we use a dual-task deep network that jointly predicts a pixel-wise segmentation map and a geometry-aware level set representation of the target. The level set representation is converted to an approximated segmentation map through a differentiable task transform layer. Simultaneously, we introduce a dual-task consistency regularization between the level set-derived segmentation maps and directly predicted segmentation maps for both labeled and unlabeled data. Extensive experiments on two public datasets show that our method can largely improve the performance by incorporating the unlabeled data. Meanwhile, our framework outperforms the state-of-the-art semi-supervised learning methods. Code is available at: <https://github.com/HiLab-git/DTC>

Introduction

Accurate and robust segmentation of organs or lesions from medical images plays an essential role in many clinical applications such as diagnosis and treatment planning (Masood et al. 2015). With a large amount of labeled data, deep learning has achieved the state-of-the-art performance on automatic image analysis (Long, Shelhamer, and Darrell 2015; Chen et al. 2018; Song et al. 2020). For medical image, however, annotations are often expensive to acquire as both expertise and time are needed to produce accurate annotations, especially in 3D volumetric images. To reduce the labeling cost, recently, many methods are proposed to develop a high-performance model for medical image segmentation

with less labeled data. For example, combining user interaction with deep neural network to perform image segmentation interactively can reduce the labeling efforts (Wang et al. 2018a,b). Self-supervised learning approaches utilize unlabeled data to train models in a supervised manner to learn fundamental knowledge for knowledge transfer (Zhu et al. 2020). Semi-supervised learning framework obtains high-quality segmentation results by learning from a limited amount of labeled data and a large set of unlabeled data directly (Li et al. 2020; Qiao et al. 2018; Zhou et al. 2019b; Xia et al. 2020; Masood et al. 2019). Weakly supervised learning methods learn from bounding boxes, scribbles or image-level tags for image segmentation rather than using pixel-wise annotation, which reduces the burden of annotation (Dai, He, and Sun 2015; Lin et al. 2016; Lee et al. 2019). In this work, we focus on the semi-supervised segmentation methods, as it is more practical to acquire a small set of fully annotated images with a large set of unannotated images.

Many recent successful SSL methods (Yu et al. 2019; Li et al. 2020; Nie et al. 2018; Li, Zhang, and He 2020) incorporate unlabeled data by performing unsupervised consistency regularization. To be specific, they can either add small perturbations to the unlabeled samples and enforce the consistency between the model predictions on the original data and the perturbed data (Yu et al. 2019; Li et al. 2020), or just directly enforce the similar prediction distributions on the entire unlabeled dataset with an adversarial regularization (Nie et al. 2018; Li, Zhang, and He 2020). Thus, we have learned that the essence of the discussed SSL works is to enforce the consistency on predictions of unlabeled data via a regularization term in loss function.

Among the aforementioned SSL works, it is delighted to see Li, Zhang, and He (2020) developed a multi-task network containing the pixel-wise and the shape-aware prediction branches, similar to previous fully supervised works (Wang et al. 2020; Xue et al. 2020). And for SSL, they consider only the shape branch to build the consistent constraints via an adversarial regularization to make prediction distributions on the entire unlabeled dataset be smooth, which still belongs to data-level regularization. We observe that various levels of information from different task branches can complement each other during training, while different focuses can lead to inherent prediction perturbation. For example, if the predictions from pixel-wise branch

基于双任务一致性的半监督医学图像分割

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摘要

基于深度学习的半监督学习（SSL）算法在医学图像分割中已取得了令人鼓舞的成果，并能够通过利用无标签数据来缓解医生昂贵的标注负担。然而，文献中大多数现有的SSL算法倾向于通过扰动网络和/或数据来对模型训练进行正则化。观察到多任务/双任务学习关注于不同层次的信息，这些信息具有固有的预测扰动，我们在本工作中提出以下问题：我们能否为SSL显式地构建任务级正则化，而不是隐式地构建网络级和/或数据级扰动并进行正则化？为了回答这个问题，我们首次提出了一种新颖的双任务一致性半监督框架。具体而言，我们使用一个双任务深度网络，联合预测目标的逐像素分割图和几何感知水平集表示。水平集表示通过可微任务变换层转换为近似的分割图。同时，我们为有标注和无标注数据引入了水平集导出的分割图与直接预测的分割图之间的双任务一致性正则化。在两个公开数据集上的大量实验表明，我们的方法通过引入无标签数据可以大幅提升性能。同时，我们的框架优于最先进的半监督学习方法。代码可在以下网址获取：

<https://github.com/HiLab-git/DTC>

引言

从医学图像中准确且稳健地分割器官或病灶，在诊断和治疗规划等许多临床应用中都发挥着至关重要的作用（Masood等，2015）。借助大量标注数据，深度学习在自动图像分析方面取得了最先进的性能（Long、Shelhamer和Darrell，2015；Chen等，2018；Song等，2020）。然而，对于医学图像而言，获取标注往往成本高昂，因为生成准确标注既需要专业知识又需要时间，尤其是在三维体图像中。为了降低标注成本，近期已有许多方法被提出，用于开发医学图像分割的高性能模型，

且所需标注数据更少。例如，将用户交互与深度神经网络相结合来交互式地执行图像分割，可以减少标注工作量（Wang等，2018a, b）。自监督学习方法利用无标注数据以监督方式训练模型，从而学习用于知识迁移的基础知识（Zhu等，2020）。半监督学习框架通过直接从有限数量的标注数据和大量无标注数据中学习，获得高质量的分割结果（Li等，2020；Qiao等，2018；Zhou等，2019b；Xia等，2020；Masood等，2019）。弱监督学习方法从边界框、涂鸦或图像级标签中学习以进行图像分割，而不是使用逐像素标注，这减轻了标注负担（Dai、He和Sun，2015；Lin等，2016；Lee等，2019）。在这项工作中，我们关注半监督分割方法，因为获取少量完全标注图像以及大量未标注图像更为实际。

许多近期成功的半监督学习方法（Yu等人，2019；Li等，2020；Nie et al. 2018；Li、Zhang和He 2020）通过执行无监督一致性正则化来利用无标签数据。具体而言，它们要么对无标注样本添加小扰动，并强制模型对原始数据和扰动数据的预测之间保持一致（Yu等人，2019；Li等，2020），要么直接通过对抗正则化在整个未标记数据集上强制相似的预测分布（Nie et al. 2018；Li、Zhang和He 2020）。因此，我们了解到所讨论的半监督学习工作的本质是通过损失函数中的正则化项来强制无标签数据预测的一致性。

在上述半监督学习工作中，令人欣喜的是，Li、Zhang和He（2020）开发了一个多任务网络，包含像素级和形状感知的预测分支，这与之之前全监督工作（Wang et al. 2020; Xue et al. 2020）类似。而在半监督学习方面，他们

仅考虑形状分支，通过对抗正则化来构建一致性约束，使整个无标注数据集上的预测分布变得平滑，这仍然属于数据级正则化。我们观察到，在训练过程中，来自不同任务分支的各种层级信息可以相互补充，而不同的关注点则可能导致固有的预测扰动。例如，如果来自像素级分支的预测

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and shape-aware branch are finally evaluated under the same criterion, we will definitely obtain different results i.e. the prediction perturbations between different tasks. Then we ask the most significant question in this work: can we explicitly build task-level regularization totally different from previous data-level regularization? Apparently the answer is yes, on the condition that the output of different task branches should be mapped/transformed to the same predefined space, where we are capable to explicitly enforce the consistency regularization between two prediction maps.

To this end, we propose a novel dual-task-consistency model for semi-supervised medical image segmentation. Our main idea is to build the consistency between a global-level level set function regression task and a pixel-wise classification task to take geometric constraints into account and utilize the unlabeled data. Our framework consists of three parts: the first part is dual-task segmentation network. Specifically, we model a segmentation problem as two different representations (tasks): predicting a pixel-wise classification map and obtaining a global-level level set function where the zero level set is the segmentation contour. We use a two-branch network to predict these two representations, and using a CNN to predict level set function is inspired by (Ma et al. 2020; Ma, He, and Yang 2020; Xue et al. 2020) to embed global information and geometric constraints into a network for better performance. The second part of this framework is a differentiable task transform layer. We use a smooth Heaviside layer (Xue et al. 2020) to convert the level set function to a segmentation probability map in a differentiable way. The third part is a combination loss function for supervised and unsupervised learning, where we design a dual-task-consistency loss function to minimize the difference between the predicted pixels-wise segmentation probability map and the probability map converted from the level set function, which can be used to boost the performance of fully supervised learning and also can be used to utilize the unlabeled data for unsupervised learning efficiently. Our proposed framework has been applied to two different semi-supervised medical image segmentation tasks: left atrium segmentation from MRI and pancreas segmentation from CT. Experimental results indicate that our proposed algorithm can improve segmentation accuracy, compared to other state-of-the-art semi-supervised segmentation methods. Overall, we present a simple yet efficient semi-supervised medical image segmentation method with dual-task consistency, which leverages the unlabeled data by encouraging consistent predictions of the same input under different tasks. Our findings during experiments include:

- 1) In the fully supervised setting, our dual-task consistency regularization outperforms the separate and joint supervision of dual tasks.
- 2) In the semi-supervised setting, the proposed framework outperforms state-of-the-art semi-supervised medical image segmentation frameworks on several clinical datasets.
- 3) Compared with existing methods, the proposed framework requires less training time and computational cost. Meanwhile, it is directly applicable to any semi-supervised medical image segmentation scene and can

easily be extended to use additional tasks given that there exists a differentiable transform between/among tasks.

Related Works

Semi-Supervised Medical Image Segmentation: For semi-supervised medical image segmentation, traditional methods mainly use hand-crafted features to design a model to perform segmentation, which includes the prior-based models (You et al. 2011) and the clustering-based models (Portela, Cavalcanti, and Ren 2014). The performance of the hand-crafted features-based models often relies on the hand-crafted features’ representation capacity. For example, the prior-based models need to design the specific prior information for different organs, which can hardly generalize to other organs. The clustering-based models are often parameter-sensitive and not robust enough, which leads to the poor prediction for objects with large shape variance.

With the ability to learn high-level semantic features automatically, deep learning has been widely used for medical image segmentation (Ronneberger, Fischer, and Brox 2015). Recently, almost all semi-supervised medical image segmentation frameworks are based on deep learning. Bai et al. (2017) developed an iterative framework where in each iteration, pseudo labels for unannotated images are predicted by the network and refined by a Conditional Random Field (CRF) (Krähenbühl and Koltun 2011), then the new pseudo labels are used to update the network. Using adversarial learning to utilize the unlabeled data is also a popular way for semi-supervised medical image segmentation. Zhang et al. (2017) proposed a new deep adversarial network (DAN) model for biomedical image segmentation by encouraging the segmentation of unannotated images to be similar to those of the annotated ones. Yu et al. (2019) extended the mean teacher model (Tarvainen and Valpola 2017) with uncertainty map guidance for semi-supervised left atrium segmentation. Li, Zhang, and He (2020) introduced a shape-aware semi-supervised segmentation strategy to leverage the unlabeled data and to enforce a geometric shape constraint on the segmentation output. Differently, our method takes advantage of geometric constraints and dual-task-consistency, which is simple yet effective for semi-supervised medical image segmentation.

Consistency Regularization: The consistency regularization plays a vital role in computer vision and image processing, especially in semi-supervised learning. For examples, Sajjadi, Javanmardi, and Tasdizen (2016) proposed a regularization with stochastic transformations and perturbations for deep semi-supervised learning, and learned from unlabeled images by minimizing the difference between the predictions of multiple passes of a training sample. Tarvainen and Valpola (2017) introduced a teacher-student consistency model to make full use of the unlabeled data, where the student model learns from the teacher model by minimizing the segmentation loss on the labeled data and the consistency loss with respect to the targets from the teacher model on all input data. Jeong et al. (2019) used consistency constraints as a tool for enhancing detection performance by making full use of available unlabeled data. Li et al. (2020) introduced a

和形状感知分支的预测最终在相同准则下进行评估，我们必然会得到不同的结果，即不同任务之间的预测扰动。于是我们提出本工作中最重要的问题：我们能否显式地构建与以往数据级正则化完全不同的**任务级正则化**？答案显然是肯定的，前提是不同任务分支的输出应被映射/变换到同一个预定义空间，在该空间中我们能够显式地强制两个预测图之间的一致性正则化。

为此，我们提出了一种新颖的双任务一致性模型，用于半监督医学图像分割。我们的核心思想是构建全局水平集函数回归任务与逐像素分类任务之间的一致性，从而将几何约束纳入考量并利用无标签数据。我们的框架由三个部分组成：**第一部分是双任务分割网络**。具体而言，我们将分割问题建模为两种不同的表示（任务）：预测逐像素分类图和获取全局水平集函数，其中零水平集即为分割轮廓。我们使用双分支网络来预测这两种表示，使用CNN预测水平集函数是受（Ma等人2020; Ma, He和Yang 2020; Xue et al. 2020）的启发，旨在将全局信息和几何约束嵌入网络中，以获得更好的性能。**该框架的第二部分是可微任务变换层**。我们使用平滑Heaviside层（Xue et al. 2020）以可微的方式将水平集函数转换为分割概率图。**第三部分是用于监督学习和无监督学习的组合损失函数**，其中我们设计了一个**双任务一致性损失函数**，以最小化预测的逐像素分割概率图与由水平集函数转换得到的概率图之间的差异，该损失函数可用于提升全监督学习的性能，也可用于高效地利用无标签数据进行无监督学习。我们提出的框架已应用于两个不同的半监督医学图像分割任务：MRI左心房分割和CT胰腺分割。实验结果表明，与其他最先进的半监督分割方法相比，我们提出的算法能够提高分割精度。总体而言，我们提出了一种简单而高效的基于双任务一致性的半监督医学图像分割方法，通过鼓励同一输入在不同任务下产生一致的预测来利用无标签数据。我们在实验中的发现包括：

1) 在全监督设置下，我们的双任务一致性正则化优于双任务的单独监督和联合监督。2) 在半监督设置下，所提出的框架在多个临床数据集上优于最先进的半监督医学图像分割框架。3) 与现有方法相比，所提出的框架需要更少的训练时间和计算成本。同时，它可直接应用于任何半监督医学图像分割场景，并且可以

在任务之间存在可微变换的情况下，轻松扩展以使用额外的任务。

相关工作

半监督医学图像分割：对于半监督医学图像分割，传统方法主要使用**手工特征**来设计模型进行分割，其中包括基于先验的模型（You等，2011）和基于聚类的模型（Portela、Cavalcanti和Ren，2014）。基于手工特征的模型其性能往往依赖于手工特征的表征能力。例如，基于先验的模型需要针对不同器官设计特定的先验信息，这很难泛化到其他器官。基于聚类的模型通常对参数敏感且不够鲁棒，导致对形状差异较大的对象预测效果不佳。

凭借自动学习高级语义特征的能力，深度学习已被广泛用于医学图像分割（Ronneberger、Fischer和Brox，2015）。近年来，几乎所有的半监督医学图像分割框架都基于深度学习。Bai等人（2017）开发了一个迭代框架，在每次迭代中，无标注图像的伪标签由网络预测并通过条件随机场（CRF）（Krähenbühl和Koltun，2011）进行细化，然后使用新的伪标签更新网络。利用对抗学习来使用无标签数据也是半监督医学图像分割的一种流行方式。Zhang等人（2017）提出了一种新的深度对抗网络（DAN）模型用于生物医学图像分割，通过鼓励无标注图像的分割结果与有标注图像的分割结果相似。Yu等人（2019）扩展了均值教师模型（Tarvainen和Valpola，2017），引入不确定性图引导用于半监督左心房分割。Li、Zhang和He（2020）引入了一种形状感知的半监督分割策略，以利用无标签数据并对分割输出施加几何形状约束。不同的是，我们的方法利用了几何约束和双任务一致性，这对于半监督医学图像分割来说简单而有效。

一致性正则化：一致性正则则在计算机视觉和图像处理中起着至关重要的作用，尤其是在半监督学习中。例如，Sajjadi、Javanmardi和Tasdizen（2016）提出了一种带有随机变换和扰动的正则化方法用于深度半监督学习，通过最小化训练样本多次前向传播预测之间的差异来从无标注图像中学习。Tarvainen和Valpola（2017）引入了一种教师-学生一致性模型以充分利用无标签数据，其中学生模型通过最小化有标注数据上的分割损失以及所有输入数据上关于教师模型目标的一致性损失来从教师模型中学习。Jeong等人（2019）将一致性约束作为提升检测性能的工具，充分利用可用的无标注数据。Li等人（2020）引入了一种

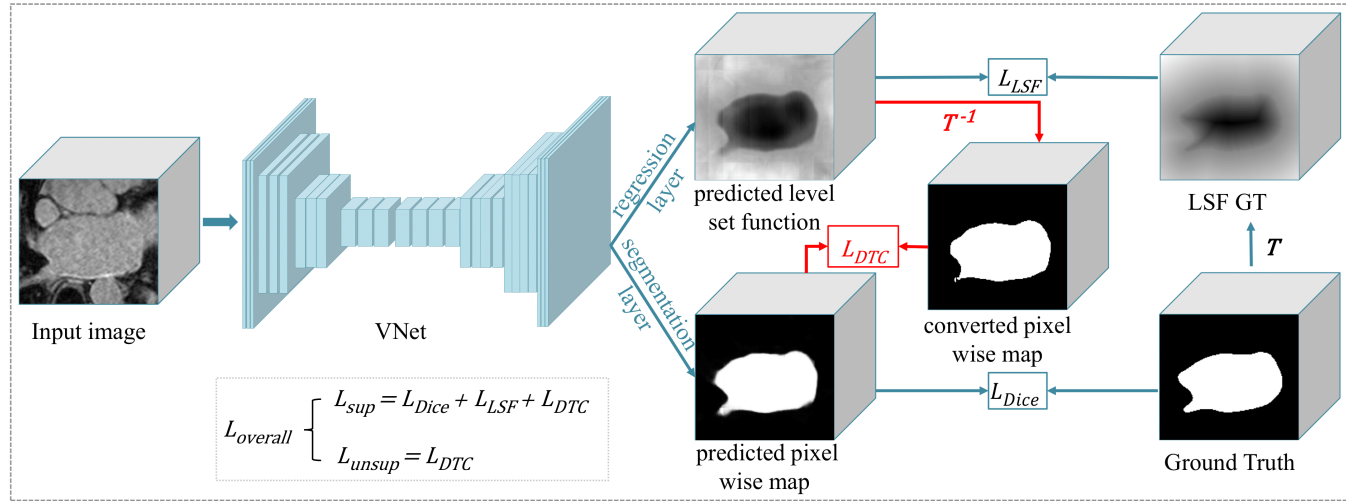


Figure 1: Overview of the proposed dual-task-consistency framework for semi-supervised medical image segmentation. The network consists of a pixel-wise classification head (task1) and a level set function regression head (task2), which employs a widely-used encoder-decoder network as the backbone, i.e., VNet (Milletari, Navab, and Ahmadi 2016). The model is optimized by minimizing supervised losses L_{Dice} , L_{LSF} on labeled data and the dual-task-consistency loss L_{DTC} on both unlabeled data and labeled data. The T function is used to transform the ground truth label map into a level set representation for supervised training. The T^{-1} function converts the level set function to a probability map to calculate the L_{DTC} .

transformation-consistent based semi-supervised segmentation method, which encourages consistent predictions of the network-in-training for the same input under different perturbations. However, these works just consider the consistency when the input under different perturbations and transformation, which ignore the consistency of different tasks. In addition, these methods need to perform forward pass two or more times for calculating the consistency loss, which increases the computational cost and running time. More recently, Zamir et al. (2020) utilized the consistency cross different tasks based on inference-path invariance, indicating it is promising to investigate task consistency. The limitation is that they require labeled data in a fully supervised manner and only studied on low-level vision tasks. In contrast to aforementioned methods, our framework aims to utilize the unlabeled data by minimizing the consistency between two tasks of a network, which considers the difference of different tasks and just needs to perform inference once. To the best of our knowledge, our work is the first to construct the task-consistency constraint for semi-supervised learning.

Methods

In this section, we introduce our proposed semi-supervised medical image framework based on dual-task-consistency. The overall framework is illustrated in Figure 1, which consists of two heads, the classification head for pixel-wise probability map and the regression head for level set representation of the target. The segmentation network takes a 3D medical image as input, and predicts the level set function and pixel-wise probability map at the same time. As a segmentation result can be represented by both a pixel-level label map and a high-level contour related to a level set function, these two predictions should be consistent for the seg-

mentation task. To utilize the unlabeled data, we propose a novel dual-task-consistency strategy, which learns from unlabeled data by minimizing the difference between the predicted pixel-wise label and the level set function. To build the consistency, a transform layer is used to convert the level set function to a pixel-wise probability map, which is implemented by smooth Heaviside function. In the following two subsections, we first introduce the dual-task consistency strategy, then introduce the semi-supervised training for segmentation through dual-task consistency.

Dual-task Consistency: In general semi-supervised learning, consistency losses are designed to encourage smooth predictions in a data-level, i.e. the predictions of same data under different transformations (Li et al. 2020) and perturbations (Ouali, Hudelot, and Tami 2020) should be the same. In contrast to data-level consistency, we enforce the task-level consistency between the pixel-level classification task, defined as task1 and the level set regression task, defined as task2. In existing works, pixel-wise classification for segmentation has been widely studied while level-set function (Li et al. 2005) is a traditional task that captures geometric active contours and distance information, which rejuvenates recently when combining with CNN (Wang et al. 2020). We introduce the level set function defined as follows:

$$\mathcal{T}(x) = \begin{cases} -\inf_{y \in \partial S} \|x - y\|_2, & x \in \mathcal{S}_{in} \\ 0, & x \in \partial S \\ +\inf_{y \in \partial S} \|x - y\|_2, & x \in \mathcal{S}_{out} \end{cases} \quad (1)$$

where x, y are two different pixels/voxels in a segmentation mask, the ∂S is the zero level set and also represents the

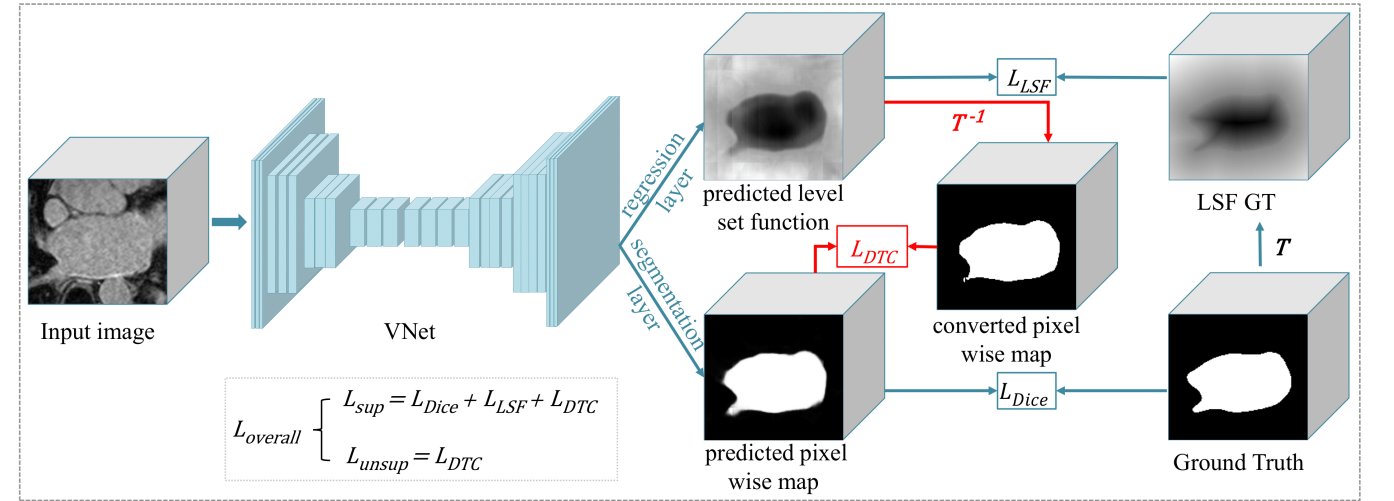


图1: 所提出的用于半监督医学图像分割的双任务一致性框架概览。该网络由一个逐像素分类头（任务1）和一个水平集函数回归头（任务2）组成，采用广泛使用的编码器-解码器网络作为主干，即VNet（Milletari, Navab和Ahmadi 2016）。该模型通过最小化有标注数据上的监督损失 L_{Dice} 、 L_{LSF} 以及无标签数据和有标注数据上的双任务一致性损失 L_{DTC} 来进行优化。 T 函数用于将真值标签图转换为水平集表示以进行监督训练。 T^{-1} 函数将水平集函数转换为概率图以计算 L_{DTC} 。

基于变换一致性的半监督分割方法，该方法鼓励训练中的网络对同一输入在不同扰动下产生一致的预测。然而，这些工作仅考虑了输入在不同扰动和变换下的一致性，忽略了不同任务之间的一致性。此外，这些方法需要执行两次或更多次前向传播来计算一致性损失，这增加了计算成本和运行时间。最近，Zamir等人（2020）基于推理路径不变性利用了跨不同任务的一致性，表明研究任务一致性具有前景。其局限性在于他们需要以全监督方式使用标注数据，并且仅在低级视觉任务上进行了研究。与上述方法不同，我们的框架旨在通过最小化网络两个任务之间的一致性来利用无标注数据，该框架考虑了不同任务的差异且只需执行一次推理。据我们所知，我们的工作为首个为半监督学习构建任务一致性约束的研究。

方法

在本节中，我们介绍所提出的基于双任务一致性的半监督医学图像框架。整体框架如图1所示，它由两个头组成：用于像素级概率图的分头头和用于目标水平集表示的回归头。分割网络以三维医学图像作为输入，同时预测水平集函数和像素级概率图。由于分割结果既可以用像素级标签图表示，也可以用与水平集函数相关的高层轮廓表示，因此这两种预测对于分割任务来说应当是一致的。

为了利用无标签数据，我们提出了一种新颖的双任务一致性策略，该策略通过最小化预测的像素级标签与水平集函数之间的差异来从无标签数据中学习。为了建立这种一致性，使用一个变换层将水平集函数转换为像素级概率图，该变换层由平滑Heaviside函数实现。在接下来的两个小节中，我们首先介绍双任务一致性策略，然后介绍通过双任务一致性进行分割的半监督训练。

双任务一致性: 在一般的半监督学习中，一致性损失旨在数据级上鼓励平滑预测，即同一数据在不同变换（Li等，2020）和扰动（Ouali, Hudelot和Tami, 2020）下的预测应当相同。与数据级一致性不同，我们在定义为任务1的像素级分类任务和定义为任务2的水平集回归任务之间强制实施任务级一致性。在现有工作中，用于分割的逐像素分类已被广泛研究，而水平集函数（Li等，2005）是一种传统任务，能够捕获几何活动轮廓和距离信息，近期在与CNN结合后重新焕发了活力（Wang等，2020）。我们引入如下定义的水平集函数：

$$\mathcal{T}(x) = \begin{cases} -\inf_{y \in \partial S} \|x - y\|_2, & x \in \mathcal{S}_{in} \\ 0, & x \in \partial S \\ +\inf_{y \in \partial S} \|x - y\|_2, & x \in \mathcal{S}_{out} \end{cases} \quad (1)$$

其中 x, y 是分割掩码中两个不同的像素/体素， ∂S 是零水平集，也表示

Algorithm 1 Semi-supervised training through Dual-task consistency,

Input: $\mathbf{x}_i \in \mathcal{D}_l + \mathcal{D}_n, \mathbf{y}_i \in \mathcal{D}_l$

Output: Dual-task model’s parameter θ_1 for segmentation head, θ_2 for level-set function (LSF) head and θ for shared-weights backbone network

- 1: $f_1(x)$ = segmentation task branch with shared parameter θ and segmentation head’s parameter θ_1
- 2: $f_2(x)$ = LSF task branch with shared parameter θ and LSF head’s parameter θ_2
- 3: **while** stopping criterion not met: **do**
- 4: Sample batch $b_l = (\mathbf{x}_i, \mathbf{y}_i) \in \mathcal{D}_l$ and $b = b_l + b_u$, where $b_u = \mathbf{x}_i \in \mathcal{D}_u$
- 5: Generating LSF ground truth $\mathcal{T}(\mathbf{y}_i)$ according to Equation. 1
- 6: Computing dual-task predictions $f_1(\mathbf{x}_i)$ and $f_2(\mathbf{x}_i), i \in \{1, \dots, N\}$ where N denotes the batch size
- 7: Applying task transform layer $\mathcal{T}^{-1}(f_2(\mathbf{x}_i))$ according to Equation. 2
- 8: $\mathcal{L}_{DTC}(\mathbf{x}) = \frac{1}{|b|} \sum_{\mathbf{x}_i \in b} \|f_1(\mathbf{x}_i) - \mathcal{T}^{-1}(f_2(\mathbf{x}_i))\|^2$
- 9: $\mathcal{L}_{LSF}(\mathbf{x}, \mathbf{y}) = \frac{1}{|b_l|} \sum_{\mathbf{x}_i, \mathbf{y}_i \in b_l} \|f_2(\mathbf{x}_i) - \mathcal{T}(\mathbf{y}_i)\|^2$
- 10: $\mathcal{L}_{Seg}(\mathbf{x}, \mathbf{y}) = 1 - \frac{1}{|b_l|} \sum_{\mathbf{x}_i, \mathbf{y}_i \in b_l} 2 \frac{\sum f_1(\mathbf{x}_i) \mathbf{y}_i}{\sum f_1(\mathbf{x}_i) + \sum \mathbf{y}_i}$
- 11: $\mathcal{L}_{total} = \mathcal{L}_{Seg} + \mathcal{L}_{LSF} + \lambda_d \mathcal{L}_{DTC}$
- 12: Computing gradient of loss function $\mathcal{L}_{total}$ and update network parameters θ_1, θ_2 and θ by back propagation.
- 13: **end while**
- 14: **return** θ_1, θ_2 and θ

contour of the target object. $\mathcal{S}_{in}$ and $\mathcal{S}_{out}$ denote the inside region and outside region of the target object. Then we define $\mathcal{T}(x)$ as the task transform from segmentation map to level-set function map in Equation. 1. To map the output of LSF task to the space of segmentation output, it is natural to think of using an inverse transform of $\mathcal{T}(x)$. However, it is impractical to integrate the exact inverse transform of $\mathcal{T}(x)$ in training due to the non-differentiability. Hence, we utilize a smooth approximation to the inverse transform of level-set function, provided that we want to guarantee the values of $\mathcal{S}_{in}$ are assigned to 1 while those of $\mathcal{S}_{out}$ are assigned to 0 in the transformed prediction map, which is defined as:

$$\mathcal{T}^{-1}(z) = \frac{1}{1 + e^{-k \cdot z}} = \sigma(k \cdot z) \quad (2)$$

where z means the level set value at pixel/voxel x . The formulation of $\mathcal{T}^{-1}(z)$ is delicate and simple as it is equal to Sigmoid function with the input multiplied by a factor k , which is selected as large as possible to approximate inverse transform of $\mathcal{T}(x)$. Thus, $\mathcal{T}^{-1}(z)$ can easily be implemented as an modified activate function followed by task2’s output. Then the differentiability can be proved as follows:

$$\begin{aligned} \frac{\partial \mathcal{T}^{-1}}{\mathrm{d}z} &= \left(\frac{1}{1 + e^{-k \cdot z}} \right)' \\ &= k \cdot \frac{1}{1 + e^{-kz}} \cdot \left(1 - \frac{1}{1 + e^{-kz}} \right) \end{aligned} \quad (3)$$

Though such approximate transform function will map the prediction space of task2 to be the same with that of task1, it naturally introduces a task-level prediction difference since task1 focuses on pixel-level reasoning while task2 attends to geometric structure information. Thus, for input $\mathbf{X}$ from a dataset $\mathcal{D}$, we define the dual-task-consistency loss $\mathcal{L}_{DTC}$ enforcing consistency between task1’s prediction $f_1(\mathbf{x}_i)$ and the transformed map of task2’s prediction $\mathcal{T}^{-1}(f_2(\mathbf{x}_i))$:

$$\begin{aligned} \mathcal{L}_{DTC}(\mathbf{x}) &= \sum_{\mathbf{x}_i \in \mathcal{D}} \|f_1(\mathbf{x}_i) - \mathcal{T}^{-1}(f_2(\mathbf{x}_i))\|^2 \\ &= \sum_{\mathbf{x}_i \in \mathcal{D}} \|f_1(\mathbf{x}_i) - \sigma(k \cdot f_2(\mathbf{x}_i))\|^2 \end{aligned} \quad (4)$$

Semi-supervised training through Dual-Task-Consistency: Let $\mathcal{D}_l$ and $\mathcal{D}_u$ be the labeled and unlabeled dataset, respectively. Let $\mathcal{D} = \mathcal{D}_l \cup \mathcal{D}_u$ be the whole provided dataset. We denote labeled data pair as $(\mathbf{X}, \mathbf{Y}) \in \mathcal{D}_l$ and unlabeled data as $\mathbf{X} \in \mathcal{D}_u$, where $\mathbf{Y}$ is groundtruth segmentation mask. We denote voxel-level pair as $(x, y) \in (\mathbf{X}, \mathbf{Y})$. For labeled data $\mathcal{D}_l$, we define the supervised loss for segmentation task as commonly used dice loss :

$$\begin{aligned} \mathcal{L}_{Seg}(\mathbf{x}, \mathbf{y}) &= \sum_{\mathbf{x}_i, \mathbf{y}_i \in \mathcal{D}_l} \mathcal{L}_{Dice}(\mathbf{x}_i, \mathbf{y}_i) \\ &= \sum_{\mathbf{x}_i, \mathbf{y}_i \in \mathcal{D}_l} \left(1 - \frac{2 \sum_{x_j \in \mathbf{x}_i, y_j \in \mathbf{y}_i} f_1(x_i) y_i}{\sum_{x_j \in \mathbf{x}_i, y_j \in \mathbf{y}_i} f_1(x_j) + \sum_{y_j \in \mathbf{y}_i} y_j} \right) \end{aligned} \quad (5)$$

where the summation for $\sum_{x_j \in \mathbf{x}_i, y_j \in \mathbf{y}_i}$ denotes voxel-wise sum in a 3D image, and the summation for $\sum_{\mathbf{x}_i, \mathbf{y}_i \in \mathcal{D}_l}$ denotes image-level sum in a dataset. Then we define the supervised loss for LSF task as $\mathcal{L}_2$ loss between the predicted probability map $f_2(\mathbf{x})$ and the transformed ground truth map $\mathcal{T}(\mathbf{y})$:

$$\mathcal{L}_{LSF}(\mathbf{x}, \mathbf{y}) = \sum_{\mathbf{x}_i, \mathbf{y}_i \in \mathcal{D}_l} \|f_2(\mathbf{x}_i) - \mathcal{T}(\mathbf{y}_i)\|^2 \quad (6)$$

It is noteworthy that for annotated images, the ground truth level set function for the LSF task can be automatically generated from labeled segmentation mask $\mathbf{Y}$ through aforementioned task transform function $\mathcal{T}$. The final loss is defined as:

$$\mathcal{L}_{total} = \mathcal{L}_{Seg} + \mathcal{L}_{LSF} + \lambda_d \mathcal{L}_{DTC} \quad (7)$$

where $\mathcal{L}_{Seg}$ and $\mathcal{L}_{LSF}$ are only used for labeled data, while $\mathcal{L}_{DTC}$ is used for both labeled and unlabeled data during training, and therefore the two tasks can jointly optimize the network with either labeled data or unlabeled data in a semi-supervised fashion. Following (Tarvainen and Valpola 2017; Yu et al. 2019), we use a time-dependent Gaussian warming up function $\lambda_d(t) = e^{(-5(1 - \frac{t}{t_{max}})^2)}$ to control the balance between the supervised loss and unsupervised consistency loss, where t denotes the current training step and t_{max} is the maximum training step. The used training algorithm for semi-supervised segmentation through dual-task consistency is shown in Algorithm. 1.

算法 1通过双任务一致性的半监督训练,

输入: $\mathbf{x}_i \in \mathcal{D}_l + \mathcal{D}_n, \mathbf{y}_i \in \mathcal{D}_l$

输出: 用于分割的双任务模型参数 θ_1

- 头, θ_2 用于水平集函数 (LSF) 头和 θ 用于共享权重主干网络
- 1: $f_1(x)$ = 具有共享参数的分割任务分支-ter θ 和分割头的参数 θ_1
- 2: $f_2(x)$ = LSF任务分支, 共享参数 θ 和 LSF头的参数 θ_2
- 3: **while** 未满足停止准则:**do**
- 4: 采样批次 $b_l = (\mathbf{x}_i, \mathbf{y}_i) \in \mathcal{D}_l$ 和 $b = b_l + b_u$, 其中 $b_u = \mathbf{x}_i \in \mathcal{D}_u$
- 5: 根据以下公式生成 LSF 真值 $\mathcal{T}(\mathbf{y}_i)$: 公式 1
- 6: 计算 双任务 预测 $f_1(\mathbf{x}_i)$ and $f_2(\mathbf{x}_i), i \in \{1, \dots, N\}$ 其中 N 表示批次大小
- 7: 根据方程 2 应用任务变换层 $\mathcal{T}^{-1}(f_2(\mathbf{x}_i))$, 根据方程 2
- 8: $\mathcal{L}_{DTC}(\mathbf{x}) = \frac{1}{|b|} \sum_{\mathbf{x}_i \in b} \|f_1(\mathbf{x}_i) - \mathcal{T}^{-1}(f_2(\mathbf{x}_i))\|^2$
- 9: $\mathcal{L}_{LSF}(\mathbf{x}, \mathbf{y}) = \frac{1}{|b_l|} \sum_{\mathbf{x}_i, \mathbf{y}_i \in b_l} \|f_2(\mathbf{x}_i) - \mathcal{T}(\mathbf{y}_i)\|^2$
- 10: $\mathcal{L}_{Seg}(\mathbf{x}, \mathbf{y}) = 1 - \frac{1}{|b_l|} \sum_{\mathbf{x}_i, \mathbf{y}_i \in b_l} 2 \frac{\sum f_1(\mathbf{x}_i) \mathbf{y}_i}{\sum f_1(\mathbf{x}_i) + \sum \mathbf{y}_i}$
- 11: $\mathcal{L}_{total} = \mathcal{L}_{Seg} + \mathcal{L}_{LSF} + \lambda_d \mathcal{L}_{DTC}$
- 12: 计算损失函数的梯度 $\mathcal{L}_{total}$ 并更新网络参数 θ_1, θ_2 和 θ , 通过反向传播
- 13: **结束循环**
- 14: **返回** θ_1, θ_2 和 θ

目标对象的轮廓。 $\mathcal{S}_{in}$ 和 $\mathcal{S}_{out}$ 分别表示目标对象的内部区域和外部区域。然后我们在公式 1 中将 $\mathcal{T}(x)$ 定义为从分割图到水平集函数图的任务变换。为了将 LSF 任务的输出映射到分割输出的空间, 自然会想到使用 $\mathcal{T}(x)$ 的逆变换。然而, 由于不可微性, 在训练中直接积分 $\mathcal{T}(x)$ 的精确逆变换是不切实际的。因此, 我们利用水平集函数逆变换的平滑近似, 前提是我们希望在变换后的预测图中保证 $\mathcal{S}_{in}$ 的值被赋为 1, 而 $\mathcal{S}_{out}$ 的值被赋为 0, 其定义如下:

$$\mathcal{T}^{-1}(z) = \frac{1}{1 + e^{-k \cdot z}} = \sigma(k \cdot z) \quad (2)$$

其中 z 表示像素/体素 x 处的水平集值。 $\mathcal{T}^{-1}(z)$ 的公式既精巧又简单, 因为它等于 Sigmoid 函数, 其输入乘以一个因子 k , 该因子被选得尽可能大, 以近似 $\mathcal{T}(x)$ 的逆变换。因此, $\mathcal{T}^{-1}(z)$ 可以很容易地作为修正激活函数实现, 并接在任务2的输出之后。然后, 可微性可以证明如下:

$$\begin{aligned} \frac{\partial \mathcal{T}^{-1}}{\mathrm{d}z} &= \left(\frac{1}{1 + e^{-k \cdot z}} \right)' \\ &= k \cdot \frac{1}{1 + e^{-kz}} \cdot \left(1 - \frac{1}{1 + e^{-kz}} \right) \end{aligned} \quad (3)$$

尽管这种近似变换函数会将任务2的预测空间映射为与任务1相同的预测空间, 但它自然会引入一个任务级预测差异, 因为任务1专注于像素级推理, 而任务2关注几何结构信息。因此, 对于来自数据集 $\mathcal{D}$ 的输入 $\mathbf{X}$, 我们定义双任务一致性损失 $\mathcal{L}_{DTC}$, 以强制任务1的预测 $f_1(\mathbf{x}_i)$ 与任务2预测的变换映射 $\mathcal{T}^{-1}(f_2(\mathbf{x}_i))$ 之间的一致性:

$$\begin{aligned} \mathcal{L}_{DTC}(\mathbf{x}) &= \sum_{\mathbf{x}_i \in \mathcal{D}} \|f_1(\mathbf{x}_i) - \mathcal{T}^{-1}(f_2(\mathbf{x}_i))\|^2 \\ &= \sum_{\mathbf{x}_i \in \mathcal{D}} \|f_1(\mathbf{x}_i) - \sigma(k \cdot f_2(\mathbf{x}_i))\|^2 \end{aligned} \quad (4)$$

通过双任务一致性的半监督训练: 设 $\mathcal{D}_l$ 和 $\mathcal{D}_u$ 分别为标注数据集和无标签数据集。设 $\mathcal{D} = \mathcal{D}_l \cup \mathcal{D}_u$ 为整个提供的数据集。我们将标注数据对表示为 $(\mathbf{X}, \mathbf{Y}) \in \mathcal{D}_l$, 无标签数据表示为 $\mathbf{X} \in \mathcal{D}_u$, 其中 $\mathbf{Y}$ 为真实分割掩膜。我们将体素级对表示为 $(x, y) \in (\mathbf{X}, \mathbf{Y})$ 。对于标注数据 $\mathcal{D}_l$, 我们将分割任务的监督损失定义为常用的 Dice 损失:

$$\begin{aligned} \mathcal{L}_{Seg}(\mathbf{x}, \mathbf{y}) &= \sum_{\mathbf{x}_i, \mathbf{y}_i \in \mathcal{D}_l} \mathcal{L}_{Dice}(\mathbf{x}_i, \mathbf{y}_i) \\ &= \sum_{\mathbf{x}_i, \mathbf{y}_i \in \mathcal{D}_l} \left(1 - \frac{2 \sum_{x_j \in \mathbf{x}_i, y_j \in \mathbf{y}_i} f_1(x_i) y_i}{\sum_{x_j \in \mathbf{x}_i, y_j \in \mathbf{y}_i} f_1(x_j) + \sum_{y_j \in \mathbf{y}_i} y_j} \right) \end{aligned} \quad (5)$$

其中, 对 $\sum_{x_j \in \mathbf{x}_i, y_j \in \mathbf{y}_i}$ 的求和表示 3D 图像中的体素级求和, 而对 $\sum_{\mathbf{x}_i, \mathbf{y}_i \in \mathcal{D}_l}$ 的求和表示数据集中的图像级求和。然后, 我们将 LSF 任务的监督损失定义为预测概率图 $f_2(\mathbf{x})$ 与变换后的真实标签图 $\mathcal{T}(\mathbf{y})$ 之间的 $\mathcal{L}_2$ 损失:

$$\mathcal{L}_{LSF}(\mathbf{x}, \mathbf{y}) = \sum_{\mathbf{x}_i, \mathbf{y}_i \in \mathcal{D}_l} \|f_2(\mathbf{x}_i) - \mathcal{T}(\mathbf{y}_i)\|^2 \quad (6)$$

值得注意的是, 对于标注图像, LSF 任务的真实标签水平集函数可以通过上述任务变换函数 $\mathcal{T}$ 从标注分割掩码 $\mathbf{Y}$ 自动生成。最终损失定义为:

$$\mathcal{L}_{total} = \mathcal{L}_{Seg} + \mathcal{L}_{LSF} + \lambda_d \mathcal{L}_{DTC} \quad (7)$$

其中 $\mathcal{L}_{Seg}$ 和 $\mathcal{L}_{LSF}$ 仅用于有标注数据, 而 $\mathcal{L}_{DTC}$ 在训练期间同时用于有标注和无标注数据, 因此这两个任务能够以半监督方式利用有标注数据或无标注数据共同优化网络。遵循 (Tarvainen 和 Valpola, 2017; Yu et al. 2019), 我们使用时间相关高斯预热函数 $\lambda_d(t) = e^{(-5(1 - \frac{t}{t_{max}})^2)}$ 来控制监督损失与无监督一致性损失之间的平衡, 其中 t 表示当前训练步, t_{max} 是最大训练步。通过双任务一致性进行半监督分割所使用的训练算法如算法 1 所示。

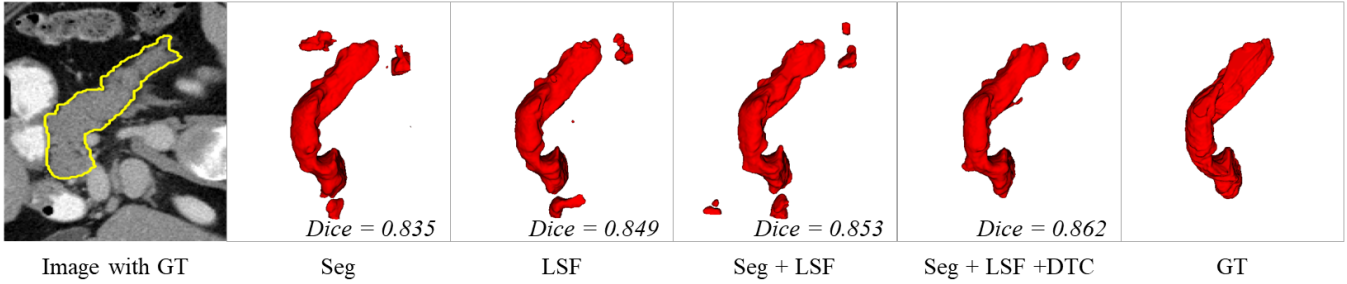


Figure 2: 3D Visualization of different training methods for pancreas segmentation. 12 annotated images without unannotated images were used for training. GT: ground truth. (best viewed in color)

Method	Scans used		Metrics				Cost	
	Labeled	Unlabeled	Dice (%)	Jaccard (%)	ASD (voxel)	95HD (voxel)	Params (M)	Training time (h)
Seg	12	0	70.63	56.72	6.29	22.54	9.44	2.1
LSF	12	0	71.78	57.55	6.31	20.74	9.44	2.1
Seg + LSF	12	0	73.08	58.65	4.47	18.04	9.44	2.2
Seg + LSF + DTC	12	0	74.84	60.78	2.17	9.34	9.44	2.3
Seg	62	0	81.78	69.65	1.34	5.13	9.44	2.3
LSF	62	0	82.25	70.23	1.18	5.19	9.44	2.5
Seg + LSF	62	0	82.46	70.61	1.22	4.97	9.44	2.5
Seg + LSF + DTC	62	0	82.80	71.05	1.45	4.67	9.44	2.5

Table 1: Ablation study of our dual-task consistency method on the Pancreas CT dataset.

Experiments and Results

Datasets and Pre-processing: To evaluate the proposed method, we apply our algorithm on two different datasets. The first is left atrial dataset (Xiong et al. 2020), which consists of 100 3D gadolinium-enhanced MR images, with a resolution of $0.625 \times 0.625 \times 0.625mm$. Following (Yu et al. 2019; Li, Zhang, and He 2020), we use 80 scans for training and 20 scans for validation, and apply the same pre-processing methods. The second is pancreas dataset (Roth et al. 2015), which includes 82 abdomen CT images. Following (Xia et al. 2020), we randomly split them into 62 images for training and 20 images for testing. In pre-processing, we use the soft tissue CT window range of $[-125, 275]$ HU (Zhou et al. 2019a), and resample all images to an isotropic resolution of $1.0 \times 1.0 \times 1.0mm$. Finally, we crop the images centering at the pancreas region based on the ground truth with enlarged margins (25 voxels) and normalize them as zero mean and unit variance. In this work, we report the performance of all methods trained with 20% labeled images and 80% unlabeled images, which is the typical semi-supervised learning experimental setting (Xia et al. 2020; Yu et al. 2019; Li, Zhang, and He 2020).

Implementation Details and Evaluation Metrics: We implement our framework in PyTorch (Paszke et al. 2019), using an NVIDIA 1080TI GPU. In this work, we use VNet (Milletari, Navab, and Ahmadi 2016) as the backbone for all experiments, and we implement dual-task VNet by adding a new regression layer at the end of the original VNet. The framework is trained by an SGD optimizer for 6000 iterations, with an initial learning rate (lr) 0.01 decayed by 0.1 every 2500 iterations. The batch size is 4, consisting of 2 labeled images and 2 unlabeled images. Following (Xue

et al. 2020), the value of k is set to 1500 in this work. We randomly crop $112 \times 112 \times 80$ (3D MRI Left Atrium) and $96 \times 96 \times 96$ (3D CT Pancreas) sub-volume as the network input. To avoid over-fitting, we use the standard on-the-fly data augmentation methods during training stage (Yu et al. 2019). Note that, in this work, the level set function is generated before the training phase rather on-the-fly, since the level set function is transform-invariant, which in result significantly speed up the training procedure. In the inference phase, we use a sliding window strategy to obtain the final results, which with a stride of $18 \times 18 \times 4$ for left atrium and $16 \times 16 \times 16$ for pancreas. At the inference time, we use the output of pixel-wise classification branch as the segmentation result. For a fair comparison, we do not use any post-processing or ensemble methods. Following (Yu et al. 2019), we use four metrics to quantitatively evaluate our method, including Dice, Jaccard, the average surface distance (ASD), and the 95% Hausdorff Distance (95HD).

The Effects of Different Tasks: To investigate the individual impact of different tasks, we first only use labeled images for training and analyze how the dual-task consistency performs when only labeled images are used. We trained the network for pancreas segmentation using the 12 labeled data and all the 62 labeled data, respectively. We compared different training strategies: 1) only using the branch for task1 (Seg), 2) only using the branch for task 2 (LSF), 3) using the two branches for task1 and task2 simultaneously (Seg + LSF), and 4) and our proposed dual-task consistency method (Seg + LSF + DTC). The performance of these variants is listed in Table. 1. It shows that the level set function regression is helpful for medical image segmentation. It also can be observed that dual-task consistency consistently im-

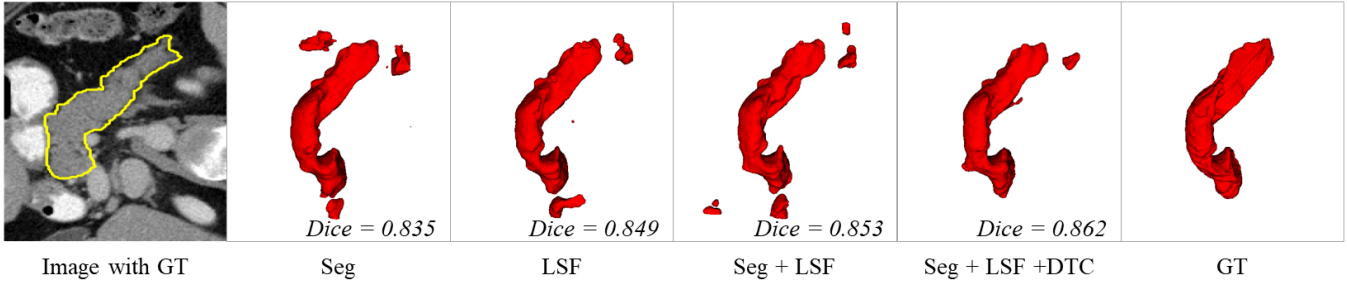


图 2: 不同训练方法用于胰腺分割的三维可视化。使用 12 张标注图像（不含未标注图像）进行训练。GT: 真实标签。（建议彩色查看）

方法	使用扫描数		指标				Cost	
	有标注	无标注	Dice (%)	Jaccard (%)	ASD(体素)	95HD(体素)	参数量(M)	训练时间(h)
Seg	12	0	70.63	56.72	6.29	22.54	9.44	2.1
LSF	12	0	71.78	57.55	6.31	20.74	9.44	2.1
分割 +水平集函数	12	0	73.08	58.65	4.47	18.04	9.44	2.2
分割 +LSF +DTC	12	0	74.84	60.78	2.17	9.34	9.44	2.3
Seg	62	0	81.78	69.65	1.34	5.13	9.44	2.3
LSF	62	0	82.25	70.23	1.18	5.19	9.44	2.5
分割 +LSF	62	0	82.46	70.61	1.22	4.97	9.44	2.5
分割 +LSF +DTC	62	0	82.80	71.05	1.45	4.67	9.44	2.5

表 1: 我们的双任务一致性方法在胰腺CT数据集上的消融研究。

实验与结果

数据集与预处理: 为评估所提出的方法，我们在两个不同的数据集上应用我们的算法。第一个是左心房数据集 (Xiong等 2020)，它由100幅3D钆增强磁共振图像组成，分辨率为 $0.625 \times 0.625 \times 0.625mm$ 。遵循(Yu et al. 2019; Li, Zhang和He 2020)，我们使用80次扫描进行训练，20次扫描进行验证，并应用相同的预处理方法。第二个是胰腺数据集(Roth et al. 2015)，包含82幅腹部CT图像。遵循(Xia等 2020)，我们将其随机划分为62幅图像用于训练，20幅图像用于测试。在预处理中，我们使用软组织CT窗范围为 $[-125, 275]$ HU (Zhou等 2019a)，并将所有图像重采样至各向同性分辨率 $1.0 \times 1.0 \times 1.0mm$ 。最后，我们基于真实标签以胰腺区域为中心裁剪图像，并扩大边界(25体素)，然后将其归一化为零均值和单位方差。在本工作中，我们报告了所有方法在20%标注图像和80%无标注图像训练下的性能，这是典型的半监督学习实验设置 (Xia等 2020; Yu et al. 2019; Li, Zhang和He 2020)。

实现细节与评估指标: 我们在PyTorch (Paszke等人, 2019) 中实现了我们的框架，使用NVIDIA 1080TI GPU。在本工作中，我们使用VNet (Milletari, Navab和Ahmadi, 2016) 作为所有实验的骨干网络，并通过在原始VNet末端添加一个新的回归层来实现双任务VNet。该框架使用SGD优化器训练6000次迭代，初始学习率 (lr) 为0.01，每2500次迭代衰减0.1。批次大小为4，由2张标注图像和2张无标注图像组成。遵循 (Xue

等人, 2020)，在本工作中 k 的值设置为1500。我们随机裁剪 $112 \times 112 \times 80$ （3D MRI左心房）和 $96 \times 96 \times 96$ （3D CT胰腺）子体积作为网络输入。为避免过拟合，我们在训练阶段使用标准的实时数据增强方法（Yu等人, 2019）。需要注意的是，在本工作中，水平集函数在训练阶段之前生成，而非实时生成，因为水平集函数具有变换不变性，从而显著加快了训练过程。在推理阶段，我们使用滑动窗口策略获得最终结果，左心房的步幅为 $18 \times 18 \times 4$ ，胰腺的步幅为 $16 \times 16 \times 16$ 。在推理时，我们使用逐像素分类分支的输出作为分割结果。为了公平比较，我们不使用任何后处理或集成方法。遵循（Yu等人, 2019），我们使用四个指标定量评估我们的方法，包括Dice、Jaccard、平均表面距离（ASD）和95%豪斯多夫距离（95HD）。

不同任务的影响: 为了研究不同任务的个体影响，我们首先仅使用标注图像进行训练，并分析在仅使用标注图像时双任务一致性的表现。我们分别使用12个标注数据和全部62个标注数据训练胰腺分割网络。我们比较了不同的训练策略: 1) 仅使用任务1 (Seg) 的分支, 2) 仅使用任务2 (LSF) 的分支, 3) 同时使用任务1和任务2的两个分支 (Seg +LSF)，以及4) 我们提出的双任务一致性方法 (Seg + LSF + DTC)。这些变体的性能列于表1中。结果表明，水平集函数回归有助于医学图像分割。还可以观察到，双任务一致性持续改

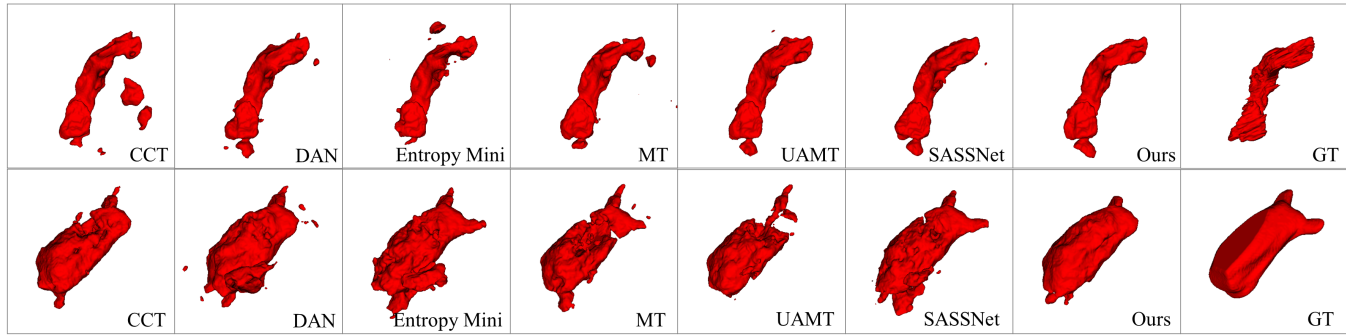


Figure 3: 3D Visualization of different semi-supervised segmentation methods under 20% labeled data (best viewed in color). The first row is a pancreas segmentation result and second row is a left atrium segmentation result.

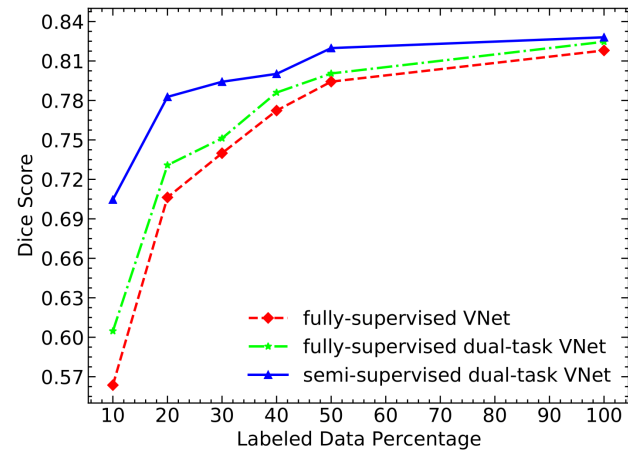


Figure 4: The pancreas segmentation performance of our semi-supervised approach with different ratio of labeled data. The dashed red and lime curves show performance of fully-supervised VNet and dual-task VNet respectively, where they were trained with only the available labeled data.

proves the performance of the dual-task VNet on 12 labeled scans and 62 labeled scans. Figure. 2 shows some visualization of different training methods, which further show the superiority of our proposed dual-task-consistency.

Effectiveness of Dual-task-Consistency for Semi-supervised Learning: Secondly, we performed a study on data utilization efficiency of our approach compared to the fully supervised VNet and dual-task VNet that only use available annotated images for training on Pancreas CT dataset. We draw the Dice score of the results in Figure.4. It can be observed that the semi-supervised method consistently performs better than the supervised approach in different labeled data settings, demonstrating that our method effectively utilizes the unlabeled data and brings performance gains. It also can be found that the performance gap between fully supervised method and semi-supervised approach narrows as more labeled images are available, which conforms to the common sense. When the number of labeled data is small, our method also can

obtain a better segmentation result than fully supervised method, indicating the promising potential of our proposed approach for further clinical use.

Comparison with Other Semi-supervised Methods: We compared our framework with six state-of-the-art semi-supervised segmentation methods, including deep adversarial network (DAN) (Zhang et al. 2017), entropy minimization approach (Entropy Mini) (Vu et al. 2019), cross-consistency training method (CCT) (Ouali, Hudelot, and Tami 2020), mean teacher self-ensembling model (MT) (Tarvainen and Valpola 2017), uncertainty-aware mean teacher model(UA-MT) (Yu et al. 2019) and shape-aware adversarial network (SASSNet) (Li, Zhang, and He 2020). Note that we used the official code and results of DAN, MT, UA-MT, SASSNet, and reimplemented the Entropy Mini and CCT for medical image segmentation, since the limitation of GPU memory, we used one main decoder and three auxiliary decoders as CCT’s implementation.

We first evaluate our proposed framework on Pancreas CT. Table. 2 shows the quantitative comparison of these methods. Compared with fully supervised VNet trained with only 12 annotated images, all semi-supervised methods taking advantages of unannotated images improve the segmentation performance significantly. The MT, UA-MT and CCT achieve slightly better performance than Entropy Mini and DAN, demonstrating that perturbation-based consistency loss is helpful for the semi-supervised segmentation problem. Moreover, the UA-MT is better than MT, since the uncertainty map can guide the student model learning efficiently. The SASSNet achieves the top performance among the existing methods, indicating the shape prior is useful for semi-supervised image segmentation. Notably, our framework achieves better performance than the state-of-the-art semi-supervised methods on all the evaluation metrics without using a complex multiple network architecture, corroborating that our dual-task-consistency has the full capability to draw out the rich information from the unlabeled data. Meanwhile, our framework does not require any multiple inference or iteratively update scheme, which reduces the computational memory cost and running time.

We further validate our proposed method on Left Atrium MRI data, which is a widely-used dataset for semi-

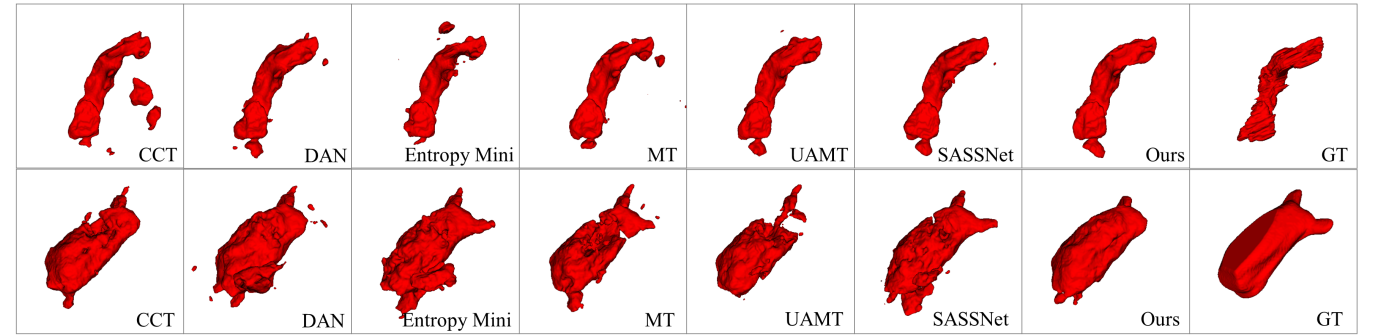


图3: d的三维可视化 在20%标注数据下不同半监督分割方法（最好以彩色查看）。第一行是胰腺分割结果，第二行是左心房分割结果。

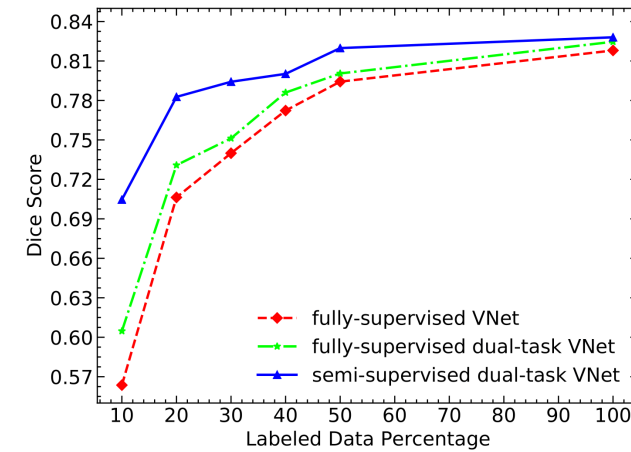


图4: 我们的半监督方法在不同标注数据比例下的胰腺分割性能。红色虚线和黄绿色曲线分别显示了全监督VNet和双任务VNet的性能，它们仅使用可用的标注数据进行训练。

善了双任务VNet在12个标注扫描和62个标注扫描上的性能。图2展示了不同训练方法的一些可视化结果，进一步表明了我们所提出的双任务一致性的优越性。

双任务一致性对半监督学习的有效性: 其次，我们针对我们的方法与仅使用可用标注图像进行训练的完全监督VNet及双任务VNet，在胰腺CT数据集上开展了数据利用效率的研究。我们在图4中绘制了结果的Dice分数。可以观察到，半监督方法在不同标注数据设置下始终优于监督方法，表明我们的方法有效利用了无标签数据并带来了性能提升。还可以发现，随着标注图像数量的增加，完全监督方法与半监督方法之间的性能差距逐渐缩小，这符合常识。当标注数据量较少时，我们的方法同样能够

获得比完全监督方法更好的分割结果，表明我们提出的方法在进一步临床应用中具有广阔前景。

与其他半监督方法的比较: 我们将我们的框架与六种最先进的半监督分割方法进行了比较，包括深度对抗网络 (DAN) (Zhang等, 2017)、熵最小化方法 (Entropy Mini) (Vu等, 2019)、交叉一致性训练方法 (CCT) (Ouali, Hudelot和Tami, 2020)、均值教师自集成模型 (MT) (Tarvainen和Valpola, 2017)、不确定性感知均值教师模型 (UA-MT) (Yu et al. 2019) 和形状感知对抗网络 (SASSNet) (Li, Zhang和He 2020)。需要注意的是，我们使用了DAN、MT、UA-MT、SASSNet的官方代码和结果，并重新实现了Entropy Mini和CCT用于医学图像分割，由于GPU内存的限制，我们使用了一个主解码器和三个辅助解码器作为CCT的实现。

我们首先在胰腺CT上评估我们提出的框架。表2展示了这些方法的定量比较。与仅使用12张标注图像训练的全监督VNet相比，所有利用未标注图像的半监督方法都显著提升了分割性能。MT、UA-MT和CCT的性能略优于Entropy Mini和DAN，表明基于扰动的一致性损失对半监督分割问题是有帮助的。此外，UA-MT优于MT，因为不确定性图能够高效地指导学生模型学习。SASSNet在现有方法中取得了最佳性能，表明形状先验对半监督图像分割是有用的。值得注意的是，我们的框架在不使用复杂的多网络架构的情况下，在所有评估指标上均优于最先进的半监督方法，证实了我们的双任务一致性具有充分的能力从无标签数据中提取丰富的信息。同时，我们的框架不需要任何多次推理或迭代更新方案，这降低了计算内存成本和运行时间。

我们进一步在左心房MRI数据上验证了我们提出的方法，该数据是半监督

Method	Scans used		Metrics				Cost	
	Labeled	Unlabeled	Dice (%)	Jaccard (%)	ASD (voxel)	95HD (voxel)	Params (M)	Training time (h)
VNet	12	0	70.63	56.72	6.29	22.54	9.44	2.1
VNet	62	0	81.78	69.65	1.34	5.13	9.44	2.3
MT (NeurIPS’ 17)	12	50	75.85	61.98	3.40	12.59	9.44	2.9
DAN (MICCAI’ 17)	12	50	76.74	63.29	2.97	11.13	12.09	3.3
Entropy Mini (CVPR’ 19)	12	50	75.31	61.73	3.88	11.72	9.44	2.2
UA-MT (MICCAI’ 19)	12	50	77.26	63.82	3.06	11.90	9.44	3.9
CCT (CVPR’ 20)	12	50	76.58	62.76	3.69	12.92	15.65	4.1
SASSNet (MICCAI’ 20)	12	50	77.66	64.08	3.05	10.93	20.46	3.9
Ours	12	50	78.27	64.75	2.25	8.36	9.44	2.5

Table 2: Quantitative comparison between our methods and other semi-supervised methods on the Pancreas CT dataset. The first and second row are our fully supervised baseline, the last row is our proposed method, others are previous methods.

Method	Scans used		Metrics				Cost	
	Labeled	Unlabeled	Dice (%)	Jaccard (%)	ASD (voxel)	95HD (voxel)	Params (M)	Training time (h)
VNet	16	0	86.03	73.26	5.75	17.93	9.44	1.8
VNet	80	0	91.14	83.32	1.52	5.75	9.44	2.0
MT(NeurIPS’ 17)	16	64	88.23	79.29	2.73	10.64	9.44	3.2
DAN (MICCAI’ 17)	16	64	87.52	78.29	2.42	9.01	12.09	3.7
Entropy Mini (CVPR’ 19)	16	64	88.45	79.51	3.72	14.14	9.44	1.9
UA-MT (MICCAI’ 19)	16	64	88.88	80.21	2.26	7.32	9.44	3.6
CCT (CVPR’ 20)	16	64	88.83	80.06	2.49	8.44	15.65	3.9
SASSNet (MICCAI’ 20)	16	64	89.27	80.82	3.13	8.83	20.46	4.4
Ours	16	64	89.42	80.98	2.10	7.32	9.44	2.2

Table 3: Quantitative comparison between our methods and other semi-supervised methods on the Left Atrium MRI dataset. The first and second row are our fully supervised baseline, the last row is our proposed method, others are previous methods.

supervised medical image segmentation (Yu et al. 2019; Li, Zhang, and He 2020). A quantitative comparison of these methods is shown in Tabel. 3. It can be found that our method achieved the best accuracy than other methods on all the evaluation metrics, especially in term of ASD and 95HD. Figure. 3 shows some visualization of pancreas segmentation and left atrium segmentation. Compared with other methods, our results have higher overlap ratio with the ground truth and produce less false positives and preserve more details, which further indicates the effectiveness, generalization and robustness of our proposed method. Furthermore, we investigated the training cost of different approaches. The quantitative comparison of network’s parameters and training time are listed in Table.2 and Table.3. It can be observed that, our framework requires less training time than MT, DAN, UAMT, CCT and SASSNet, since our framework use a simple network with fewer parameters and does not need to pass an image many times in an iteration. Compared with Entropy Mini and fully supervised baseline, our method achieved better accuracy with comparable computational cost. Thus, our experiments prove that our method attains the best accuracy, networks’ parameters and computational-cost trade-offs.

Discussion and Conclusion

In this paper, we have presented a novel and simple semi-supervised medical image segmentation framework through dual-task consistency, which is a task-level consistency-based framework for semi-supervised segmentation. We use

a dual-task network that simultaneously predicts a pixel-level classification map and a level set representation of the segmentation that is able to capture global-level shape and geometric information. In order to build a semi-supervised training framework, we enforce dual-task consistency between classification map prediction and LSF prediction via a task-transform layer. We achieve stat-of-the-art results on two 3D medical image datasets including left atrial dataset in MR scans and pancreas dataset in CT scans. The superior performance demonstrates the effectiveness, robustness and generalization of our proposed framework. In this work, we focus on single-class segmentation to simplify the presentation. However, our method extends to the multi-class case in a straightforward manner.

In addition, our proposed method can easily be extended to use additional tasks such as edge extraction (Zhen et al. 2020) and key-points estimation (Cheng et al. 2020) as long as there exists differentiable transform between two tasks. We also hope to inspire the whole computer vision community, as it is possible to construct tasks consistency in a semi-supervised fashion in many directions such as two-stream video recognition (Simonyan and Zisserman 2014), multi-task image reconstruction (Zamir et al. 2018, 2020) etc. to leverage a large amount of unlabeled data. In the future, we will extend this method to more computer vision applications to reduce labeling efforts and further investigate the fusion strategy to ensemble all different tasks’ prediction results for better performance.

方法	使用扫描数		指标				Cost	
	有标注	无标注	Dice (%)	Jaccard (%)	ASD(体素)	95HD(体素)	参数量(M)	训练时间(h)
VNet	12	0	70.63	56.72	6.29	22.54	9.44	2.1
VNet	62	0	81.78	69.65	1.34	5.13	9.44	2.3
MT(NeurIPS’ 17)	12	50	75.85	61.98	3.40	12.59	9.44	2.9
DAN (MICCAI’ 17)	12	50	76.74	63.29	2.97	11.13	12.09	3.3
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UA-MT (MICCAI’ 19)	12	50	77.26	63.82	3.06	11.90	9.44	3.9
CCT (CVPR’ 20)	12	50	76.58	62.76	3.69	12.92	15.65	4.1
SASSNet (MICCAI’ 20)	12	50	77.66	64.08	3.05	10.93	20.46	3.9
Ours	12	50	78.27	64.75	2.25	8.36	9.44	2.5

表2：我们的方法与其他半监督方法在胰腺CT数据集上的定量比较。第一行和第二行是我们的全监督基线，最后一行是我们提出的方法，其余为先前的方法。

方法	使用扫描数		指标				Cost	
	有标注	无标注	Dice (%)	Jaccard (%)	ASD(体素)	95HD(体素)	参数量(M)	训练时间(h)
VNet	16	0	86.03	73.26	5.75	17.93	9.44	1.8
VNet	80	0	91.14	83.32	1.52	5.75	9.44	2.0
MT(NeurIPS’ 17)	16	64	88.23	79.29	2.73	10.64	9.44	3.2
DAN (MICCAI’ 17)	16	64	87.52	78.29	2.42	9.01	12.09	3.7
Entropy Mini (CVPR’ 19)	16	64	88.45	79.51	3.72	14.14	9.44	1.9
UA-MT (MICCAI’ 19)	16	64	88.88	80.21	2.26	7.32	9.44	3.6
CCT (CVPR’ 20)	16	64	88.83	80.06	2.49	8.44	15.65	3.9
SASSNet (MICCAI’ 20)	16	64	89.27	80.82	3.13	8.83	20.46	4.4
Ours	16	64	89.42	80.98	2.10	7.32	9.44	2.2

表3：我们的方法与其他半监督方法在左心房MRI数据集上的定量比较。第一行和第二行为我们的全监督基线，最后一行为我们提出的方法，其他为先前方法。

医学图像分割中广泛使用的数据集（Yu et al. 2019; Li, Zhang和He 2020）。这些方法的定量比较如表3所示。可以发现，我们的方法在所有评估指标上都取得了比其他方法更好的准确率，尤其是在ASD和95HD方面。图3展示了胰腺分割和左心房分割的一些可视化结果。与其他方法相比，我们的结果与真实标签具有更高的重叠率，产生更少的假阳性并保留了更多细节，这进一步表明了我们所提出方法的有效性、泛化性和鲁棒性。此外，我们研究了不同方法的训练成本。网络参数和训练时间的定量比较列于表2和表3中。可以观察到，我们的框架比MT、DAN、UAMT、CCT和SASSNet所需的训练时间更少，因为我们的框架使用了一个参数更少的简单网络，并且不需要在每次迭代中将图像多次传递。与Entropy Mini和全监督基线相比，我们的方法在可比的计算成本下取得了更好的准确率。因此，我们的实验证明，我们的方法在准确率、网络参数和计算成本之间取得了最佳权衡。

讨论与结论

在本文中，我们提出了一种新颖且简单的基于双任务一致性的半监督医学图像分割框架，这是一种基于任务级一致性的半监督分割框架。我们使用

一个双任务网络，同时预测像素级分类图和分割的水平集表示，该表示能够捕获全局级形状与几何信息。为了构建半监督训练框架，我们通过任务变换层在分类图预测和LSF预测之间强制施加双任务一致性。我们在两个3D医学图像数据集上取得了最先进的结果，包括MR扫描中的左心房数据集和CT扫描中的胰腺数据集。优越的性能证明了我们所提出框架的有效性、鲁棒性和泛化能力。在这项工作中，我们专注于单类分割以简化表述。然而，我们的方法可以直接扩展到多类情况。

此外，只要两个任务之间存在可微变换，我们提出的方法可以很容易地扩展到使用额外任务，例如边缘提取（Zhen等人，2020）和关键点估计（Cheng等人，2020）。我们也希望激励整个计算机视觉社区，因为在许多方向上都可以以半监督的方式构建任务一致性，例如双流视频识别（Simonyan和Zisserman, 2014）、多任务图像重建（Zamir等人，2018, 2020）等，以利用大量的无标签数据。未来，我们将把该方法扩展到更多的计算机视觉应用中，以减少标注工作量，并进一步研究融合策略，以集成所有不同任务的预测结果，从而获得更好的性能。

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